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NEW RESEARCH HORIZON ***Review***

The human sperm epigenome and its potential role in embryonic development

Douglas T. Carrell[1] and Saher Sue Hammoud

IVF and Andrology Laboratories, Departments of Surgery (Urology), Obstetrics and Gynecology, and Physiology, 675 S. Arapeen Dr, Suite #205, University of Utah School of Medicine, Salt Lake City, UT 84108, USA

[1]Correspondence address. Tel: +1-801-587-3772; Fax: +1-801-581-6127; E-mail: douglas.carrell@hsc.utah.edu

ABSTRACT: Along with many of the genome-wide transitions in chromatin composition throughout spermatogenesis, epigenetic modifications on histone tails and DNA are continuously modified to ensure stage specific gene expression in the maturing spermatid. Recent findings have suggested that the repertoire of epigenetic modifications in the mature sperm may have a potential role in the developing embryo and alterations in the epigenetic profile have been associated with infertility. These changes include DNA demethylation and the retention of modified histones at important developmental, signaling and micro-RNA genes, which resemble the epigenetic state of an embryonic stem cell. This review assesses the significance of epigenetic changes during spermatogenesis, and provides insight on recent associations made between altered epigenetic profiles in the mature sperm and its relationship to infertility.

Key words: methylation / histone / *in vitro* fertilization / embryogenesis / epigenetic

Introduction

In vitro fertilization (IVF) and other assisted reproductive technologies (ART) have accounted for ~3 million births since the world's first IVF baby was born in 1978 (Cohen, 1978). Reports examining the long-term health consequences of these babies are limited; however, follow-up studies have reported increased intrauterine growth restriction and lower birth weights in singletons conceived by IVF compared with natural conceptions (Steel and Sutcliffe, 2009). More recently, increased perinatal mortality, congenital anomalies and epigenetic abnormalities have been reported to be associated with IVF (Seif *et al*., 2006; Allen *et al*., 2008; Kalra and Molinaro, 2008; Reefhuis *et al*., 2009; Steel and Sutcliffe, 2009).

The underlying causes of increased anomalies in IVF offspring are unknown, but alterations in the normal epigenetic state of gametes of severely infertile patients undergoing IVF has been proposed as one potential contributor (Cutfield *et al*., 2007; Lim *et al*., 2009; Manipalviratn *et al*., 2009). Epigenetic modifications are covalent modifications present on either the DNA itself or to the proteins that are closely associated with DNA (histones in somatic cells and histones and protamines in sperm), both of which are important in modifying gene expression without changing the genetic code itself. These modifications comprise what is commonly referred to as the epigenome, which in somatic cells regulates cellular fate and function (Bernstein *et al*., 2002, 2005, 2006; Li, 2002; Jones and Baylin, 2007). It is now well understood that the epigenome can become disrupted or altered, which may contribute significantly to the onset of epigenetic changes observed in many diseases and may be causative of some diseases (Feinberg, 2007; Jones and Baylin, 2007).

Recent studies have demonstrated that sperm have unique and potentially important epigenetic modifications. This brief review describes chromatin and epigenetic changes throughout spermatogenesis, their potential role in normal embryonic development, and their implications in male infertility.

Histone Modifications During Spermatogenesis

Male germ cells undergo unique and extensive chromatin and epigenetic remodeling soon after their specification (determination to become a spermatocyte) and during the differentiation process to become a mature spermatozoon (Seki *et al*., 2005). Although the mechanisms regulating and orchestrating specification and spermiogenesis remain poorly understood, some progress has been made in elucidating the changes associated with the complex cellular changes. During mitosis and meiosis, male germ cell DNA is packaged in nucleosomes, comprised of histone 2A (H2A), histone 2B (H2B), histone 3 (H3) and histone 4 (H4), all of which are susceptible to covalent modifications, such as methylation, acetylation, ubiquitination

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and phosphorylation. Each of these chemical modifications to histones works alone or in concert to influence gene repression and/or activation (Fig. 1).

Histone methylation on lysine (K) residues of H3 or H4 can promote gene activation and/or repression (Lachner and Jenuwein, 2002; Suganuma and Workman, 2008). Monomethylation, dimethylation and trimethylation modifications of H3K4, H3K9 or H3K27 display tightly controlled temporal expression and ensure proper progression through spermatogenesis (Khalil *et al*., 2004; Godmann *et al*., 2007; Payne and Braun, 2006). The level of H3K4 methylation peaks in the spermatogonial stem cell stage (Fig. 2), and a targeted loss of H3K4 methylation, caused by reduction of Mll2 activity (an H3K4 methyl transferase), results in a dramatic reduction in the number of spermatocytes (Table I), suggesting that H3K4 methylation is essential

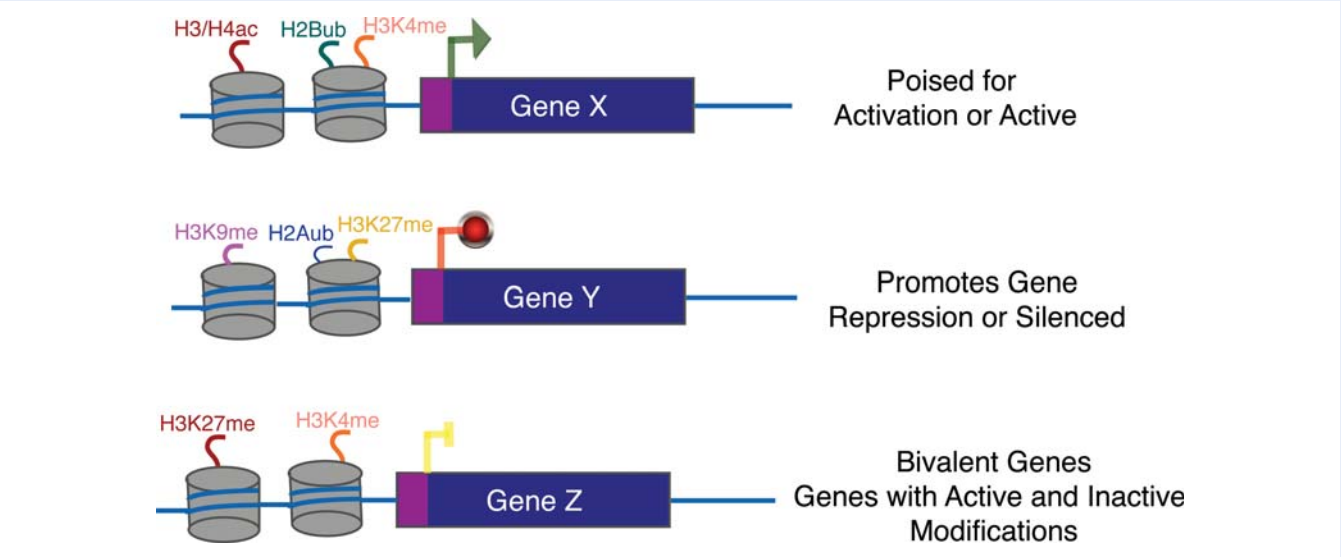


Figure 1 Chromatin modifications determine gene state. Histone modifications promote either gene activation or repression; however, in embryonic stem cells and sperm a subset of genes are commonly associated with both active and inactive marks. ac = acetylation, me = methylation, ub = ubiquitination.

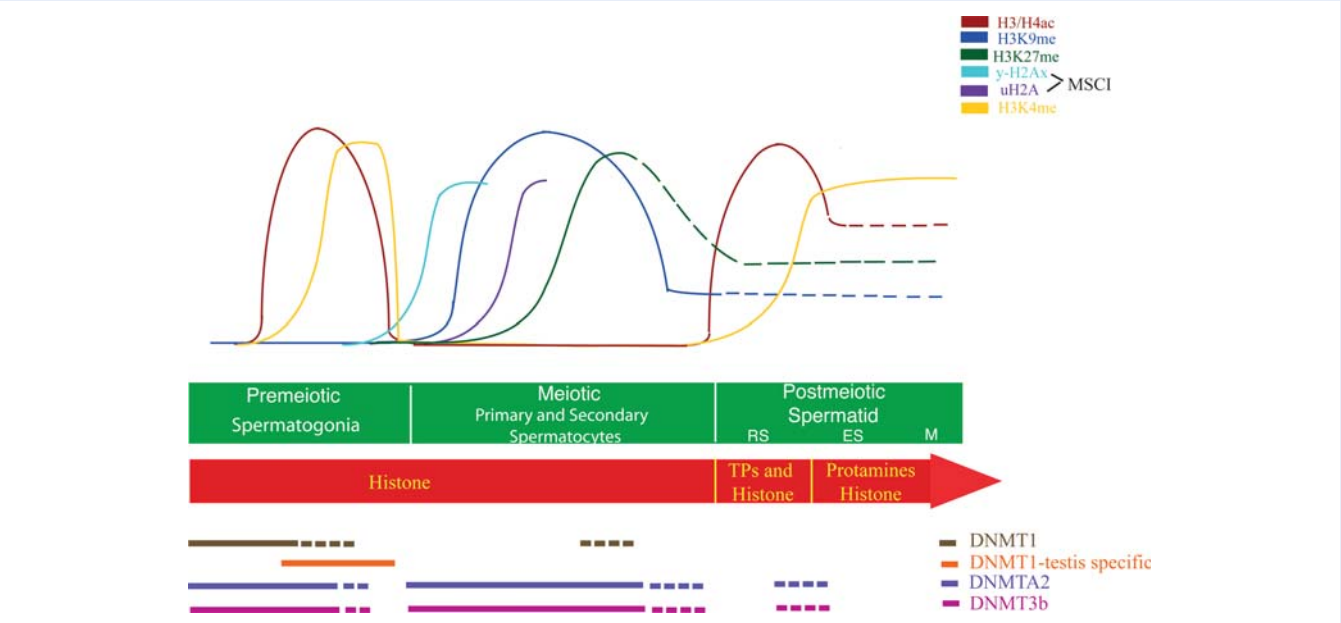


Figure 2 Hypothetical expression profiles of histone modifications and DNA methyltransferases during spermatogenesis. In the top panel, solid lines indicate investigated histone level. In the bottom panel dashed lines indicate low expression. Ac = acetylation, me = methylation, TPs = transition proteins, Ax = phosphorylation, MSC1 = meiotic sex chromosome inactivation, DNMT = DNA methyl transferase, RS = round spermatid, ES = elongated spermatid, M = mature spermatid.

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Table I A Summary of chromatin modifiers and their associated functions

Chromatin modifier	Function
DNMT1	Maintenance DNA methyltransferase
DNMT3a	*de-novo* DNA methyltransferase
DNMT3b	*de-novo* DNA methyltransferase
DNMT3L	*de-novo* DNA methyltransferase (no catalytic activity)
MII	H3K4 methyl transferase
JHDM2A	H3K9 demethylase
HAT	Histone acetyl transferase
HDACs	Histone deacetylase
LSD1/KDM	H3K4 demethylase

for the exit from the stem cell stage and commitment to become a spermatocyte (Glaser *et al*., 2009). In contrast, H3K9 methylation and H3K27methylation are low in the stem cell and increase during meiosis (Fig. 2), persisting long after meiosis is complete, presumably to ensure gene-silencing (Payne and Braun, 2006). The role ascribed to each of these modifications has been primarily characterized by immunofluorescence data with no gene specific localization. Methylation on lysine 9 of histone H3 (H3K9me) is associated with the sex chromosomes, euchromatin and heterochromatin in the late pachytene stage, however, the levels of H3K9 methylation drop upon completion of meiosis. This reduction in H3K9me is concurrently associated with an increase in H3K4me levels (Fig. 2) (Glaser *et al*., 2009).

The timing of establishment and removal of methylation marks is critical to normal spermatogenesis, as demonstrated by numerous transgenic animal models. Loss of LSD1/KDM1 (H3K4me demethylase) during mid to late meiosis in *Caenorhabditis elegans* results in germ cell apoptosis and progressive sterility that is maintained through many generations (Shi *et al*., 2004; Lee *et al*., 2005; Katz *et al*., 2009) (Table I). Similarly, removal of H3K9me at the end of meiosis is essential for the completion of spermatogenesis (Okada *et al*., 2007). Targeted disruption of the H3K9 demethylase JHDM2A (JmjC domain containing histone demethylase 2A, also known as JMJD1A) (Table I) results in complete loss of protamine 1 (PRM1) and transition protein 1 (TNP1) expression, defective chromatin condensation, and infertility (Okada *et al*., 2007). These studies show that methylation acts through various mechanisms to guide spermatogenesis (Table I).

Histone acetylation of lysine residues is dynamically regulated by histone acetyltransferases (HATs) and histone deacetylases (HDACs), both of which are shown to be essential for spermatogenesis (Christensen *et al*., 1984; Grimes and Henderson, 1984; Hazzouri *et al*., 2000; Lahn *et al*., 2002; Sonnack *et al*., 2002; Fenic *et al*., 2004, 2008; An, 2007, Kurtz *et al*., 2007). Histone acetylation relaxes chromatin and promotes polymerase II (Pol II) gene transcription, whereas deacetylation is associated with gene silencing (Jenuwein and Allis, 2001). Acetylation levels on both H3 and H4 are high in the stem cell phase and are completely removed during meiosis (Fig. 2). Re-acetylation of H4 happens in the elongating spermatid and is known to be a prerequisite for the histone-to-protamine exchange process (Fig. 2) (Hazzouri *et al*., 2000). However, recent studies show that both H3 and H4 acetylation occur simultaneously in the elongating stage, possibly extending to H2A and H2B acetylation as well (Nair *et al*., 2008). These data raise the question whether all histones may need to be acetylated to ensure proper histone to protamine exchange in maturing sperm. Enzymes involved in H4 hyperacetylation in the round spermatid are unknown, however, two candidates have emerged, testis specific chromodomain protein (CDY) and HAT (monocytic leukemia) 4 (MYST4). Both of these acetyltransferases are expressed during the maturing spermatid stage, localize to the nucleus, and have been shown to have potent H4 acetylase activity (Lahn *et al*., 2002; McGraw *et al*., 2007). H3 acetylation in the elongating spermatid was shown to be Pygopus 2 (Pygo 2) dependent. Pygo 2 has an evolutionarily conserved plant homeodomain (PHD) finger domain that binds trimethylated H3K4 and facilitates H3 acetylation (Nair *et al*., 2008). Evidence for H2B acetylation has been recently described by mass spectrometery studies, however, very little is known about its function during spermatogenesis or the enzymes required for its acetylation (Lu *et al*., 2009).

Although acetylation is broad, studies using HDAC inhibitors have demonstrated that the acetylation process is also necessarily specific. The effect of HDAC inhibitors on spermatogenesis is poorly defined, but a few studies have shown that although treating mice with HDAC inhibitors did not result in hyperacetylation, it did cause severe infertility (Fenic *et al*., 2004). Trichostatin A (TSA) treated animals had no evidence of H4 hypercetylation in the round spermatid, but the number of spermatids was significantly reduced (Fenic *et al*., 2004, 2008). The inability to detect the hyperacetylation following TSA treatment maybe due to an increase in apoptosis in cells with abnormal acetylation levels, or due to a compensatory mechanism involving alternative HDACs that are insensitive to TSA (Pivot-Pajot *et al*., 2003).

Histone phosphorylation occurs at serine residues of all core histones and is generally associated with gene activation (Berger, 2002). However, H2Ax phosphorylation (also known as γH2Ax) in germ cells confers the formation of X/Y sex body during spermatogenesis and is a marker for telomere clustering and double stranded breaks (Fernandez-Capetillo *et al*., 2003a, b). H2Ax phosphorylation is dependent on the ataxia telangiectasia DNA repair and Rad3 related protein ATR, and on the tumor suppressor BRCA1. Together γH2Ax, ATR and BRCA1 initiate meiotic sex chromosome inactivation (MSCI), but to maintain MSCI throughout the pachytene stage there are many other epigenetic modifications including ubiquinated H2A that are localized to the XY body, however, the exact function each performs are unknown (Hoyer-Fender, 2003).

The effect of ubiquination varies depending on the core histone modified: ubiquination of H2A associates with transcriptional repression, whereas, mono-ubiquination of H2B is linked to transcriptional activation in sperm (Baarends *et al*., 2005; Zhu *et al*., 2005). In male germ cells, recruitment of ubiquinated H2A to the sex body and telomeres occurs long after γH2Ax incorporation (Fig. 2), which indicates that H2A-ubiquination may be involved in maintaining silencing in the inactive chromatin, but not establishing MSCI.

These brief descriptions of histone modifications during spermatogenesis (data summarized in Fig. 2) demonstrate the varied

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ways in which epigenetic modification regulate spermatogenesis. Although most histones are replaced with protamines during the elongating spermatid stage, some of the modified nucleosomes escape the histone to protamine transition and as a result are retained in mature sperm, suggesting that these retained nucleosomes may also play a role in the paternal contribution to the embryo.

The Role of Paternal Histones in the Epigenetic Control of Embryogenesis

A hallmark of spermiogenesis is the widespread changes in chromatin structure during spermiogenesis, including the exchange of most canonical histones for protamines (Ward and Coffey, 1991). Protamines are small basic proteins that bind DNA to form toroids; tightly packed structures which compact the genome beyond what is attainable by nucleosomes. The high-level of compaction is an essential attribute for genome transport in the mature sperm head (Balhorn *et al*., 2000). The histone to protamine exchange process is incomplete, with a small percentage (5–15%) of the genome bound to nucleosomes (Tanphaichitr *et al*., 1978; Wykes and Krawetz, 2003). The replacement of somatic histones by protamines is important for nuclear chromatin compaction, sperm maturation and fertility (Gatewood *et al*., 1987, 1990; Balhorn *et al*., 1988; de Yebra *et al*., 1998; Corzett *et al*., 2002; Aoki *et al*., 2004, 2005, 2006a, b, c; Hammoud *et al*., 2009a, b). In humans the relative proportion of protamine-1 (P1) to protamine-2 (P2) is strictly regulated at approximately a 1:1 ratio and alterations in the P1/P2 ratio are very rare in fertile men and relatively common in infertile men (Chevaillier *et al*., 1987; Balhorn *et al*., 1988, 1999; Belokopytova *et al*., 1993; de Yebra *et al*., 1998; Carrell and Liu, 2001; Corzett *et al*., 2002; Aoki *et al*., 2005, Oliva, 2006; Carrell *et al*., 2007). Reports from many labs have shown that changes in the P1/P2 ratio are not only associated with altered sperm quality, but also associated with decreased embryo quality and IVF outcome compared with infertile patients with a normal P1/P2 ratio (Aoki *et al*., 2006a, b, c; Depa-Martynow *et al*., 2007). These preliminary associations suggest that protamines and histones may have a greater role during the preimplantation embryo development than previously expected.

The retained nucleosomes are comprised of canonical histones (H2A, H2B, H3, H4) as well as a testes-specific histone variant (tH2B) (Gatewood *et al*., 1990; Kimmins and Sassone-Corsi, 2005). Until recently, the role for the retained nucleosomes was unknown, but it was speculated that the retained nucleosomes were either remnants of incomplete histone to protamine replacement, or that they may have a biologically significant role during early embryogenesis. In recent work from our laboratory, we have shown that nucleosomes retained in sperm are not simply randomly distributed remnants of inefficient protamine replacement, but are instead significantly enriched at many loci important for embryo development, including genes of key embryonic transcription factors and signaling pathway proteins. Histones were also significantly enriched at the promoters of miRNAs and imprinted genes (addressed in more detail below) (Hammoud *et al*., 2009a, b). These findings challenge the widely assumed notion that the paternal genome provides little in epigenetic contributions beyond a small set of paternally imprinted genes and a modest repertoire of packaged RNA, due to the repackaging of the vast majority of the genome by protamine.

The identification of retained nucleosomes at key developmental genes was striking, but to have any potential paternal contribution to the developing embryo secondary modifications on the retained nucleosomes (such as the modification discussed above) may be key to differentiate the paternally poised genes from all other genes that have acquired acetylated histones following protamine displacement soon after fertilization. To better understand the chromatin landscape at histone-associated developmental, signaling and miRNA genes three chromatin attributes were tested: histone variants, histone modifications and DNA methylation. We hypothesized that consistent, orderly and biologically relevant patterns in histone modifications, variants and/or DNA methylation could imply a programmatic marking, such as 'poising' genes for activation during early embryogenesis, as opposed to a random and non-biologically relevant inefficiency in protamine replacement and epigenetic marking.

The first logical candidate tested for paternal genome poising was the testis specific H2B variant (tH2B) which is incorporated late in spermatogenesis and comprises a large percentage of retained histones (Hammoud *et al*., 2009a, b). Analysis of tH2B distribution throughout the sperm genome revealed a very significant enrichment of this histone variant at genes for ion channels and genes involved in spermiogenesis, but not at promoters for developmental genes. A second variant in spermatogenesis that has been recently implicated in gene poising in other cell types, such as embryonic stem (ES) cells, is the histone variant H2Az. H2Az in ES cells were shown to be a key regulator of chromatin function and associated with targets of the Polycomb complex at genes essential for ES cell differentiation (Creyghton *et al*., 2008). In contrast to its role in ES cells, H2Az in sperm was enriched at pericentric heterochromatin, which is consistent with prior immunostaining studies (Rangasamy *et al*., 2003). Data from these two variants suggested that canonical histones might be an alternative at developmental genes.

Chromatin immunoprecipitation (ChIP) followed by either microarray or deep sequencing analysis clearly shows that modified canonical nucleosomes reveal attributes of a remnant spermatogenesis program (primarily enriched with H3K4me3), as well as a future developmental program (enriched with H3K27me3, H3K4me2 and H3K4me3) (Hammoud *et al*., 2009a, b). GO term analyses for H3K4me3 yielded genes important for changing nuclear architecture, RNA metabolism, spermatogenesis and a selected number of transcription factors important for embryonic development. However, the majority of the developmental and signaling transcription factors were significantly enriched with H3K27me3 and H3K4me2. Interestingly, many of the developmental promoters harboring an activation mark such as H3K4me3 also retained a silencing mark H3K27me3 (Bernstein *et al*., 2002, 2006), resembling the bivalently marked (H3K4me3 and H3K27me3) promoters commonly seen in ES cells that are typically silenced prior to ES cell differentiation, but necessary for embryonic differentiation (Fig. 1). Furthermore, modified canonical nucleosomes extended beyond known gene promoters to promoters of non-coding RNAs, miRNAs and imprinted loci. No clear pattern was seen at many of the miRNA and non-coding RNA promoters, but this is primarily due to the limited knowledge on the functional role of these miRNAs in development. Taken together, these data reveal extensive

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histone modification patterns, and significant similarities to patterns observed in ES cells that may indicate a significant role for sperm epigenetic marking in the establishment of embryonic totipotency.

Significance of DNA methylation in the paternal germline

Packaging and transcriptional control of DNA in eukaryotes is by in large governed by the highly conserved role of histones, however, in higher organisms DNA methylation has been shown to have an essential role in normal embryonic development, regulating gene expression, X chromosome inactivation, genomic imprinting and silencing of endogenous retroviruses (Jaenisch and Jahner, 1984; Surani, 1998; Ng and Bird, 1999). DNA methylation occurs primarily at cytosine residues in a CpG context and is catalyzed by two important classes of DNA methyltransferases (DNMTs): maintenance DNA methyltaransferase (*Dnmt1*) and *de novo* DNA methyltransferases (*Dnmt3a*, *Dnmt3b*, *Dnmt3l*) (Fig. 2, Table I) (Eden and Cedar, 1994).

Dnmt1 is the major methyltransferase in somatic cells. It has a preference for hemimethylated DNA and is critical for the maintenance of DNA methylation patterns in replicating cells (Bestor, 1992; Lei *et al*., 1996) (Table I). Mice homozygous for a targeted partial deletion of *Dnmt1* or complete loss of function of *Dnmt1* have retarded growth and die by mid-gestation (Li *et al*., 1992, 1996). *Dnmt1* deficient embryos have less than 5% of the normal levels of cytosine methylation, regain biallelic expression at imprinted loci and have ectopic expression of *Xist* and retrotransposons (Li *et al*., 1993; Panning and Jaenisch, 1996; Walsh *et al*., 1998; Goll and Bestor, 2005).

The second class of DNMTs are *de novo* methyltransferases (*Dnmt3a*, *3b*, *3l*) and are essential for establishing new DNA methylation patterns during development (Table I). Embryos lacking *Dnmt3a* or *3b* lost all *de novo* methylation capabilities in ES cells and early embryos (resulted in embryonic lethality), but had no effect on the maintenance of imprinted loci (Hsieh, 1999; Lyko *et al*., 1999; Okano *et al*., 1999). Dnmt3a and b were required for methylating centromeric loci and imprinted genes. *Dnmt3l* is closely related in sequence to *DNMT3A* and *B*, but lacks the catalytic domain. *Dnmt3l* mediates *de novo* methylation by stimulating the catalytic activity of *DNMT3A2*, an isoform variant of DNMT3A (Hata *et al*., 2002; Suetake *et al*., 2004). *Dnmt3A2* along with its cofactor *Dnmt3l* establishes locus-specific DNA methylation of paternal imprints prior to meiosis in spermatogenesis (Bourc'his *et al*., 2001; Kaneda *et al*., 2004). Males' haploinsufficient for *Dnmt3l* are phenotypically normal and fertile, but have subtle changes in methylation and chromatin state of the genome in pre-meiotic spermatogonia (Bourc'his and Bestor, 2004; Webster *et al*., 2005). The phenotype for DNMT3L null male germ cells is significantly different in male versus female germ cells. Male germ cells fail to methylate LINE-1 (long interspersed elements) and IAP (intracisternal A particles) classes of retrotransposons, have severe asynapsis at meiotic prophase, and undergo apoptosis of all germ cells before pachytene (Bourc'his and Bestor, 2004). Whereas, methylation patterns at the small number of paternally methylated DMRs are almost normal, suggesting some functional redundancy between DNMTs (Bourc'his and Bestor, 2004; Webster *et al*., 2005). Strikingly, in the oocyte DNMT3L deficient germ cells carried out normal meiosis and methylation at repeat sequences, but the obvious methylation defect was limited to maternally imprinted loci. The differences observed between male and female knockouts are intriguing and raise many questions regarding targeting mechanism and differential regulation between sexes.

In addition to the DNMT gene knockout or haploinsufficiency studies used to assess the functional significance of DNA methylation in the germ line and early embryo, animals treated with DNMT inhibitors such as 5-aza-2'-deoxyCytidine showed altered gene expression patterns and loss of methylation in the germline. The severity of the phenotype observed in the treatment group was heavily dependent on the duration of treatment: short-term exposure in mice and rats decreased fertility (Seifertova *et al*., 1976; Raman and Narayan, 1995),whereas, prolonged treatment of 5-azacytidine (11weeks) in male rats resulted in a dose-dependent reductions in testis, epididymal weights and sperm counts, increase in germ cell apoptosis, and a significant increase in preimplantation loss (Doerksen and Trasler, 1996; Doerksen *et al*., 2000). These findings suggest that sperm DNA methylation plays a critical role in the differentiation of spermatogonia and early embryo viability, however, this appears counterintuitive since DNA methylation patterns are erased and reestablished immediately after fertilization and once again when primordial germ cells (PGCs) reach the genital ridge (Reik *et al*., 2001; Hajkova *et al*., 2002, Seki *et al*., 2005, 2007).

In humans, the relationship between bulk methylation levels with respect to IVF outcome was recently examined in one study. No significant correlation was made between bulk DNA methylation levels and the fertilization rate or embryo quality, but a lower 5-methyl cytosine signal (<555 AU) intensity correlated with a lower pregnancy (8.3 versus 33.3%) rates (Benchaib *et al*., 2005). These preliminary associations in humans are interesting, but interpretation and implications of measuring bulk methylation levels are limited and provide very little understanding of causality or the programs perturbed (activation of retrotransposons, changes at imprinted genes, etc.), and more genome-wide approaches are needed.

Sperm DNA Methylation Profiles and its Role in the Paternal Germline

Germ cells undergo extensive epigenetic reprogramming during proliferation and migration to the genital ridge (Seki *et al*., 2005, 2007). DNA methylation is erased and re-established in a sex and sequence specific manner during gametogenesis (Reik *et al*., 2001; Hajkova *et al*., 2002). The timing for methylation reestablishment differs between sexes and is continual for certain gene classes (Trasler, 2006). In males, *de novo* genomic methylation begins prenatally (prospermatogonia) at imprinted loci and repetitive elements with a general consensus being, that methylation patterns are completed by the end of pachytene stage of meiosis (Oakes *et al*., 2007a, b). However, exceptions have been reported at a few gene promoters that are expressed early in spermatogenesis (Pgk-2, ApoA1 and Oct-3/4) but are silenced in the maturing spermatid by gradually acquiring methylation in postmeiotic spermatocytes (Ariel *et al*., 1991, 1994).

Recent genome-wide methylation studies have indicated that the sperm epigenome differs markedly from that of somatic cells, but is very similar to ES cells and embryonic germ cells (EG) (Eckhardt *et al*., 2006, Oakes *et al*., 2007a, b; Weber *et al*., 2007; Farthing

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et al., 2008). Examining sperm DNA methylation of chromosomes 6, 20 and 22 using restriction landmark genomic studies showed that many loci were differentially methylated between sperm and somatic cells. Extending methylation analysis genome-wide to all promoters of the human genome showed that the promoters that were differentially methylated in sperm and somatic tissue (or acquire methylation upon differentiation) were promoters with 'weak' CpG islands. More recently, we have shown in our gene ontology analysis that the hypomethylated promoters in the mature sperm are the promoters of developmental transcription and signaling factors. Interestingly, the DNA hypomethylated promoters in mature sperm greatly overlapped with the developmental promoters bound by the self renewal network transcription factors in human ES cells (e.g. OCT4, SOX2, NANOG, KLF4 and FOXD3 proteins) (Boyer *et al*., 2005). In ES cells, these pluripotency proteins promote self renewal and also work with repressive polycomb complexes (PRC2) to help repress a large set of developmental regulators (including *HOX* genes) to prevent differentiation (Cao *et al*., 2002; Bernstein *et al*., 2006; Lee *et al*., 2006; Muller and Kassis, 2006; Schwartz *et al*., 2006; Takahashi *et al*., 2007; Tanay *et al*., 2007; Wernig *et al*., 2007; Kopp *et al*., 2008). This overlap suggested that pluripotency or polycomb complex factors might be involved in the establishment and maintenance of sperm DNA methylation poising in sperm. Unfortunately, pluripotency and polycomb factors were not detected in the mature sperm, in fact many of the pluripotency promoters including several key members of the self-renewal network (*OCT4*, *NANOG*, *FOXD3*) themselves acquire methylation throughout spermatogenesis, whereas their developmental target genes remain hypomethylated, consistent with recent studies in mice (Down *et al*., 2008; Farthing *et al*., 2008; Illingworth *et al*., 2008; Mohn *et al*., 2008). These findings show that genes encoding early developmental transcription factors as well as signaling proteins are DNA hypomethylated and histone bound. Furthermore, developmental promoters are selectively methylated during development, which may help commit cells to differentiation decisions. Histone retention and DNA demethylation contribute to a poised state that ensures transcriptional competence and activation of developmental regulators in the early embryo.

Epigenetic alterations and male infertility

The incidence of infertility has been rising, currently affecting one in every seven couples in the western population. Male infertility is responsible for roughly half of the cases of infertility. The underlying cause of male infertility is unexplained in ~50% infertile men, and genetic causes have been proposed to be likely (Carrell, 2008a, b; Matzuk and Lamb, 2008). Several studies have explored possible genetic causes using mouse models (O'Bryan and de Kretser, 2006), candidate gene sequencing (Miyamoto *et al*., 2003; Aoki *et al*., 2006a, b, c; Hammoud *et al*., 2007, 2009a, b), and recently the first genome-wide association study in oligozoospermic and azoospermic men has been completed (Aston and Carrell, 2009). However, these studies have revealed that single gene polymorphisms are not likely to be the cause of most cases of male infertility, but male infertility is likely to be a multifactorial disease. Similar to other complex diseases, such as cancer, epigenetic alterations may be a component contributing to infertility.

As described above, recent studies have shown that epigenetic modifications in sperm (both histone modifications and DNA methylation) appear to poise the paternal genome to participate in early embryogenesis. Additionally, several studies indicate that DNA methylation is altered, in at least imprinted genes, oligozoospermic men and men with improper histone to protamine replacement (Marques *et al*., 2004, 2008; Bowdin *et al*., 2007; Doornbos *et al*., 2007; Hammoud, in press; Kobayashi *et al*., 2007, 2009). These observations beg the question of whether methylation defects, of both imprinted and non-imprinted genes, as well as other epigenetic defects (such as histone localization or modifications in the mature sperm), may play an important role in the development and growth of ART offspring (Manipalviratn *et al*., 2009). If epigenetic profiles of the mature sperm are critical, then alterations in epigenetic patterns in infertile males can provide a logic for the increased risk for preterm birth, low birthweight, congenital anomalies, perinatal mortality, and several other pregnancy-related complications seen at a higher frequency in babies conceived by IVF (Hansen *et al*., 2002; Kalra and Molinaro, 2008). This growing field of epigenetics in early gametes and embryos may be of benefit in understanding such observations (Carrell, 2008a, b). Current studies in our laboratory are focusing on genome-wide changes in histone localization as well as DNA methylation (discussed below) in male partners of recurrent pregnancy loss patients, repeated failed IVF patients and infertile males with an altered histone to protamine ratio.

Current findings showing that epigenetics patterns in germline are extensive and of potential significance, only strengthens further the previous associations that showed abnormal methylation of imprinted genes (genes expressed in a parent-of-origin manner), in the gametes of some infertile men or babies conceived by IVF (DeBaun *et al*., 2003; Gosden *et al*., 2003; Maher, 2005). A gain or loss of expression of imprinted genes has been implicated in many diseases (Jaenisch and Bird, 2003; Seitz *et al*., 2004; Morgan *et al*., 2005; Royo *et al*., 2006) including Beckwith–Weidemann syndrome (BWS) and Angelman's Syndrome (AS), both of which have been significantly correlated with IVF babies (DeBaun *et al*., 2003; Gosden *et al*., 2003; Maher, 2005).

Whether the increased incidence of imprinting abnormalities in IVF babies arises from *in vitro* manipulations of embryos or gametes, are due to ovulation induction medications, or are inherited from the gametes of infertile patients is unclear (Cummins and Jequier, 1994; de Kretser, 1995; Edwards and Ludwig, 2003; Marques *et al*., 2004, 2008; Bowdin *et al*., 2007, Doornbos *et al*., 2007, Kobayashi *et al*., 2007, 2009; Laprise, 2009). Support for all former hypotheses have been provided but this review will focus on one aspect, preexistent methylation alterations in the gametes of infertility patients (Cummins and Jequier, 1994; de Kretser, 1995; Edwards and Ludwig, 2003; Marques *et al*., 2004, 2008; Bowdin *et al*., 2007, Doornbos *et al*., 2007, Kobayashi *et al*., 2007, 2009; Laprise, 2009). This finding first surfaced a few years ago and showed that DNA methylation patterns at paternally imprinted loci are altered in the sperm of severely oligozoospermic patients (Marques *et al*., 2004, 2008). Subsequently in 2007, Houshdaran *et al*. reported that a broader alteration in DNA methylation in sperm is seen at a handful of imprinted loci, CpG islands upstream of gene promoters, and a few repetitive elements in infertile patients with poor semen parameters (Houshdaran *et al*., 2007). Furthermore, our lab has shown

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that methylation alterations extend beyond severely oligozoospermic patients to patients with relatively normal sperm counts but with abnormal chromatin packaging, defined by an altered P1/P2 ratio (Hammoud, in press). Interestingly, in the small number of patients tested with either oligozoospermia or abnormal protamine expression we observed that methylation alterations varied between the two different etiologies of infertility. For instance oligozoospermic patients were hypermethylated at *MEST*, an imprinted gene associated with Silver Russell Syndrome (SRS), whereas abnormal protamine patients had significant changes at *LIT1* and small nuclear ribonucleoprotein polypeptide N (*SNRPN*), genes that may be associated with cases of transient neonatal diabetes milletus (TNDM) and alternative splicing (AS). These findings suggest that the risk of transmitting epigenetic alterations may vary with the classification of infertility; however, it is important to note that not all patients or alleles were affected to a similar extent. The differences in the degree of methylation within some genes or alleles compared with others raises an important questions for future studies: whether there is a variable risk to the different CpGs and whether abnormal methylation has a threshold level for conferring disease risk in the embryo or is a gradual continuum.

Pre-existent methylation alterations in the gametes of infertile patients pose a risk for transgenerational epigenetic inheritance. Evidence for transgenerational epigenetic inheritance remains controversial in humans (Oswald *et al*., 2000; Reik *et al*., 2003; Morgan *et al*., 2005), however, considerable evidence for transgenerational epigenetic inheritance in mice has become apparent at intracisternal A particles (IAPs) and imprinted genes(Bultman *et al*., 1992; Morgan *et al*., 1999; Rakyan *et al*., 2003). In humans, evidence for germline epigenetic inheritance has come almost exclusively from epidemiological studies. The strongest evidence for germline epigenetic inheritance comes from the work of Horsthemke and colleagues (Buiting *et al*., 2003) where they have shown that the presence of epimutations and not genetic mutations, at the *SNRPN–SNURF upstream reading frame* locus was inherited from the paternal grandmother (Buiting *et al*., 2003). Furthermore, Kagami *et al*. (2007) showed that defective methylation at the DMR of *MEST* in sperm may have been inherited by an ART born baby with SRS. More recently, Kobayashi *et al*. showed abnormal DNA methylation at many imprinted loci in 17 of 78 assisted reproductive technology (ART) embryos (21.8%) tested. Although some of the imprinting errors identified may have risen during the ART process, however, in seven cases hypomethylation at H19 and GTL2 was present both in the sperm and in the embryo, suggesting that abnormal hypomethylation may be paternally inherited (Kobayashi *et al*., 2009). In summary, human transgenerational epigenetic inheritance is uncertain, but is possible if altered DNA methylation is inefficiently cleared between generations or if methylation pattern are not properly reestablished due to refractory elements such as retrotransposons or certain histone modifications in the embryo and PGCs.

Structural and *in vitro* data show that certain modified nucleosmes such as H3K4 methylation can deter DNA methylation in mice (Ooi *et al*., 2007). This interdependency or cooperativity relationship can have potential implications in reprogramming, especially at imprinted loci. Recent studies have shown that maternally and paternally imprinted alleles retain differential histone modifications (methylation and acetylation) to promote either allele activation or repression in somatic cells (Fournier *et al*., 2002; Delaval and Feil, 2004; Delaval *et al*., 2007). Consistent with the findings in somatic cells, imprinted genes (imprints are established in the gametes) in human sperm used similar poising mechanisms: H3K4me3 associated with many of the paternally expressed DMRs (Hammoud *et al*., 2009a, b), whereas, maternally imprinted (paternally repressed) loci lacked H3K4me3 and had moderate levels of H3K9me3, a repressive chromatin signature, residing at a few tested loci by qPCR. The presence of modified nucleosomes in the germline may serve as an epigenetic cellular memory to help reestablish and maintain parent of origin identity. However, in the cases of male infertility with altered histone retention this may be problematic at the time of reprogramming if the retained nucleosomes in the mature sperm are improperly placed or modified. Whether this is one of the underlying factors that contributes to the poor embryo outcome in patients with an abnormal histone to protamine ratio is unknown.

Conclusion and future directions

Sperm chromatin state is highly dynamic and retains important chromatin attributes that help facilitate the proper progression of spermatogenesis as well as being a potential contributor to early developmental processes. This continual epigenetic remodeling state may make sperm cells susceptible to impediments of environmental factors, aging process, or diseases such as infertility, but the ramifications of the altered chromatin states in the germ-line are not entirely known. Future studies are needed to establish perdurance of paternally retained modified nucleosomes in the early embryo, and their potential effects if abnormally retained.

References

Allen C, Bowdin S, Harrison RF, Sutcliffe AG, Brueton L, Kirby G, Kirkman-Brown J, Barrett C, Reardon W, Maher E. Pregnancy and perinatal outcomes after assisted reproduction: a comparative study. *Ir J Med Sci* 2008;**177**:233–241.

An W. Histone acetylation and methylation: combinatorial players for transcriptional regulation. *Subcell Biochem* 2007;**41**:351–369.

Aoki VW, Wilcox AL, Thorp C, Hamilton BD, Carrell DT. Improved *in vitro* fertilization embryo quality and pregnancy rates with intracytoplasmic sperm injection of sperm from fresh testicular biopsy samples vs. frozen biopsy samples. *Fertil Steril* 2004;**82**:1532–1535.

Aoki VW, Liu L, Carrell DT. Identification and evaluation of a novel sperm protamine abnormality in a population of infertile males. *Hum Reprod* 2005;**20**:1298–1306.

Aoki VW, Christensen GL, Atkins JF, Carrell DT. Identification of novel polymorphisms in the nuclear protein genes and their relationship with human sperm protamine deficiency and severe male infertility. *Fertil Steril* 2006a;**86**:1416–1422.

Aoki VW, Emery BR, Liu L, Carrell DT. Protamine levels vary between individual sperm cells of infertile human males and correlate with viability and DNA integrity. *J Androl* 2006b;**27**:890–898.

Aoki VW, Liu L, Jones KP, Hatasaka HH, Gibson M, Peterson CM, Carrell DT. Sperm protamine 1/protamine 2 ratios are related to *in vitro* fertilization pregnancy rates and predictive of fertilization ability. *Fertil Steril* 2006c;**86**:1408–1415.

Ariel M, McCarrey J, Cedar H. Methylation patterns of testis-specific genes. *Proc Natl Acad Sci USA* 1991;**88**:2317–2321.

Ariel M, Cedar H, McCarrey J. Developmental changes in methylation of spermatogenesis-specific genes include reprogramming in the epididymis. *Nat Genet* 1994;**7**:59–63.

Downloaded from http://molehr.oxfordjournals.org/ at Carleton University on June 22, 2014

Aston KI, Carrell DT. Genome-wide Study of Single Nucleotide Polymorphisms Associated with Azoospermia and Severe Oligozoospermia. *J Androl* 2009;**30**:711–725.

Baarends WM, Wassenaar E, van der Laan R, Hoogerbrugge J, Sleddens-Linkels E, Hoeijmakers JH, de Boer P, Grootegoed JA. Silencing of unpaired chromatin and histone H2A ubiquitination in mammalian meiosis. *Mol Cell Biol* 2005;**25**:1041–1053.

Balhorn R, Reed S, Tanphaichitr N. Aberrant protamine 1/protamine 2 ratios in sperm of infertile human males. *Experientia* 1988;**44**:52–55.

Balhorn R, Cosman M, Thornton K, Krishnan VV, Corzett M, Bench G, Kramer C, Lee J, Hud NV, Allen MJ *et al.* Protamine mediated condensation of DNA in mammalian sperm. In Gagnon C (ed). *The Male Gamete: From Basic Science to Clinical Applications*. Vienna, IL: Cache River Press, 1999, 55–70.

Balhorn R, Brewer L, Corzett M. DNA condensation by protamine and arginine-rich peptides: analysis of toroid stability using single DNA molecules. *Mol Reprod Dev* 2000;**56**:230–234.

Belokopytova IA, Kostyleva EI, Tomilin AN, Vorob'ev VI. Human male infertility may be due to a decrease of the protamine P2 content in sperm chromatin. *Mol Reprod Dev* 1993;**34**:53–57.

Benchaib M, Braun V, Ressnikof D, Lornage J, Durand P, Niveleau A, Guerin JF. Influence of global sperm DNA methylation on IVF results. *Hum Reprod* 2005;**20**:768–773.

Berger SL. Histone modifications in transcriptional regulation. *Curr Opin Genet Dev* 2002;**12**:142–148.

Bernstein BE, Humphrey EL, Erlich RL, Schneider R, Bouman P, Liu JS, Kouzarides T, Schreiber SL. Methylation of histone H3 Lys 4 in coding regions of active genes. *Proc Natl Acad Sci USA* 2002; **99**:8695–8700.

Bernstein BE, Kamal M, Lindblad-Toh K, Bekiranov S, Bailey DK, Huebert DJ, McMahon S, Karlsson EK, Kulbokas EJ 3rd, Gingeras TR *et al.* Genomic maps and comparative analysis of histone modifications in human and mouse. *Cell* 2005;**120**:169–181.

Bernstein BE, Mikkelsen TS, Xie X, Kamal M, Huebert DJ, Cuff J, Fry B, Meissner A, Wernig M, Plath K *et al.* A bivalent chromatin structure marks key developmental genes in embryonic stem cells. *Cell* 2006; **125**:315–326.

Bestor TH. Activation of mammalian DNA methyltransferase by cleavage of a Zn binding regulatory domain. *EMBO J* 1992;**11**:2611–2617.

Bourc'his D, Bestor TH. Meiotic catastrophe and retrotransposon reactivation in male germ cells lacking Dnmt3L. *Nature* 2004; **431**:96–99.

Bourc'his D, Xu GL, Lin CS, Bollman B, Bestor TH. Dnmt3L and the establishment of maternal genomic imprints. *Science* 2001; **294**:2536–2539.

Bowdin S, Allen C, Kirby G, Brueton L, Afnan M, Barratt C, Kirkman-Brown J, Harrison R, Maher ER, Reardon W. A survey of assisted reproductive technology births and imprinting disorders. *Hum Reprod* 2007;**22**:3237–3240.

Boyer LA, Lee TI, Cole MF, Johnstone SE, Levine SS, Zucker JP, Guenther MG, Kumar RM, Murray HL, Jenner RG *et al.* Core transcriptional regulatory circuitry in human embryonic stem cells. *Cell* 2005;**122**:947–956.

Buiting K, Gross S, Lich C, Gillessen-Kaesbach G, el-Maarri O, Horsthemke B. Epimutations in Prader-Willi and Angelman syndromes: a molecular study of 136 patients with an imprinting defect. *Am J Hum Genet* 2003;**72**:571–577.

Bultman SJ, Michaud EJ, Woychik RP. Molecular characterization of the mouse agouti locus. *Cell* 1992;**71**:1195–1204.

Cao R, Wang L, Wang H, Xia L, Erdjument-Bromage H, Tempst P, Jones RS, Zhang Y. Role of histone H3 lysine 27 methylation in Polycomb-group silencing. *Science* 2002;**298**:1039–1043.

Carrell DT, Liu L. Altered protamine 2 expression is uncommon in donors of known fertility, but common among men with poor fertilizing capacity, and may reflect other abnormalities of spermiogenesis. *J Androl* 2001;**22**:604–610.

Carrell DT, Emery BR, Hammoud S. Altered protamine expression and diminished spermatogenesis: what is the link? *Hum Reprod Update* 2007;**13**:313–327.

Carrell DT. Contributions of spermatozoa to embryogenesis: assays to evaluate their genetic and epigenetic fitness. *Reprod Biomed Online* 2008a;**16**:474–484.

Carrell DT. Elucidating the genetics of male infertility: understanding transcriptional and translational regulatory networks involved in spermatogenesis. *Int J Androl* 2008b;**31**:455–456.

Chevaillier P, Mauro N, Feneux D, Jouannet P, David G. Anomalous protein complement of sperm nuclei in some infertile men. *Lancet* 1987;**2**:806–807.

Christensen ME, Rattner JB, Dixon GH. Hyperacetylation of histone H4 promotes chromatin decondensation prior to histone replacement by protamines during spermatogenesis in rainbow trout. *Nucleic Acids Res* 1984;**12**:4575–4592.

Cohen ME. The 'brave new baby' and the law: fashioning remedies for the victims of *in vitro* fertilization. *Am J Law Med* 1978;**4**:319–336.

Corzett M, Mazrimas J, Balhorn R. Protamine 1: protamine 2 stoichiometry in the sperm of eutherian mammals. *Mol Reprod Dev* 2002;**61**:519–527.

Creyghton MP, Markoulaki S, Levine SS, Hanna J, Lodato MA, Sha K, Young RA, Jaenisch R, Boyer LA. H2AZ is enriched at polycomb complex target genes in ES cells and is necessary for lineage commitment. *Cell* 2008;**135**:649–661.

Cummins JM, Jequier AM. Treating male infertility needs more clinical andrology, not less. *Hum Reprod* 1994;**9**:1214–1219.

Cutfield WS, Hofman PL, Mitchell M, Morison IM. Could epigenetics play a role in the developmental origins of health and disease? *Pediatr Res* 2007; **61**:68R–75R.

de Kretser DM. The potential of intracytoplasmic sperm injection (ICSI) to transmit genetic defects causing male infertility. *Reprod Fertil Dev* 1995; **7**:137–141; discussion 141–2.

de Yebra L, Ballesca JL, Vanrell JA, Corzett M, Balhorn R, Oliva R. Detection of P2 precursors in the sperm cells of infertile patients who have reduced protamine P2 levels. *Fertil Steril* 1998;**69**:755–759.

DeBaun MR, Niemitz EL, Feinberg AP. Association of *in vitro* fertilization with Beckwith–Wiedemann syndrome and epigenetic alterations of LIT1 and H19. *Am J Hum Genet* 2003;**72**:156–160.

Delaval K, Feil R. Epigenetic regulation of mammalian genomic imprinting. *Curr Opin Genet Dev* 2004;**14**:188–195.

Delaval K, Govin J, Cerqueira F, Rousseaux S, Khochbin S, Feil R. Differential histone modifications mark mouse imprinting control regions during spermatogenesis. *EMBO J* 2007;**26**:720–729.

Depa-Martynow M, Kempisty B, Lianeri M, Jagodzinski PP, Jedrzejczak P. Association between fertilin beta, protamines 1 and 2 and spermatid-specific linker histone H1-like protein mRNA levels, fertilization ability of human spermatozoa, and quality of preimplantation embryos. *Folia Histochem Cytobiol* 2007;**45**:S79–S85.

Doerksen T, Trasler JM. Developmental exposure of male germ cells to 5-azacytidine results in abnormal preimplantation development in rats. *Biol Reprod* 1996;**55**:1155–1162.

Doerksen T, Benoit G, Trasler JM. Deoxyribonucleic acid hypomethylation of male germ cells by mitotic and meiotic exposure to 5-azacytidine is associated with altered testicular histology. *Endocrinology* 2000; **141**:3235–3244.

Doornbos ME, Maas SM, McDonnell J, Vermeiden JP, Hennekam RC. Infertility, assisted reproduction technologies and imprinting disturbances: a Dutch study. *Hum Reprod* 2007;**22**:2476–2480.

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Down TA, Rakyan VK, Turner DJ, Flicek P, Li H, Kulesha E, Graf S, Johnson N, Herrero J, Tomazou EM *et al*. A Bayesian deconvolution strategy for immunoprecipitation-based DNA methylome analysis. *Nat Biotechnol* 2008;**26**:779–785.

Eckhardt F, Lewin J, Cortese R, Rakyan VK, Attwood J, Burger M, Burton J, Cox TV, Davies R, Down TA *et al*. DNA methylation profiling of human chromosomes 6, 20 and 22. *Nat Genet* 2006;**38**:1378–1385.

Eden S, Cedar H. Role of DNA methylation in the regulation of transcription. *Curr Opin Genet Dev* 1994;**4**:255–259.

Edwards RG, Ludwig M. Are major defects in children conceived *in vitro* due to innate problems in patients or to induced genetic damage? *Reprod Biomed Online* 2003;**7**:131–138.

Farthing CR, Ficz G, Ng RK, Chan CF, Andrews S, Dean W, Hemberger M, Reik W. Global mapping of DNA methylation in mouse promoters reveals epigenetic reprogramming of pluripotency genes. *PLoS Genet* 2008;**4**:e1000116.

Feinberg AP. Phenotypic plasticity and the epigenetics of human disease. *Nature* 2007;**447**:433–440.

Fenic I, Sonnack V, Failing K, Bergmann M, Steger K. *In vivo* effects of histone-deacetylase inhibitor trichostatin-A on murine spermatogenesis. *J Androl* 2004;**25**:811–818.

Fenic I, Hossain HM, Sonnack V, Tchatalbachev S, Thierer F, Trapp J, Failing K, Edler KS, Bergmann M, Jung M *et al*. In vivo application of histone deacetylase inhibitor trichostatin-a impairs murine male meiosis. *J Androl* 2008;**29**:172–185.

Fernandez-Capetillo O, Liebe B, Scherthan H, Nussenzweig A. H2AX regulates meiotic telomere clustering. *J Cell Biol* 2003a;**163**:15–20.

Fernandez-Capetillo O, Mahadevaiah SK, Celeste A, Romanienko PJ, Camerini-Otero RD, Bonner WM, Manova K, Burgoyne P, Nussenzweig A. H2AX is required for chromatin remodeling and inactivation of sex chromosomes in male mouse meiosis. *Dev Cell* 2003b;**4**:497–508.

Fournier C, Goto Y, Ballestar E, Delaval K, Hever AM, Esteller M, Feil R. Allele-specific histone lysine methylation marks regulatory regions at imprinted mouse genes. *EMBO J* 2002;**21**:6560–6570.

Gatewood JM, Cook GR, Balhorn R, Bradbury EM, Schmid CW. Sequence-specific packaging of DNA in human sperm chromatin. *Science* 1987;**236**:962–964.

Gatewood JM, Cook GR, Balhorn R, Schmid CW, Bradbury EM. Isolation of four core histones from human sperm chromatin representing a minor subset of somatic histones. *J Biol Chem* 1990;**265**:20662–20666.

Glaser S, Lubitz S, Loveland KL, Ohbo K, Robb L, Schwenk F, Seibler J, Roellig D, Kranz A, Anastassiadis K *et al*. The histone 3 lysine 4 methyltransferase, Mll2, is only required briefly in development and spermatogenesis. *Epigenetics Chromatin* 2009;**2**:5.

Godmann M, Auger V, Ferraroni-Aguiar V, Di Sauro A, Sette C, Behr R, Kimmins S. Dynamic regulation of histone H3 methylation at lysine 4 in mammalian spermatogenesis. *Biol Reprod* 2007;**77**:754–764.

Goll MG, Bestor TH. Eukaryotic cytosine methyltransferases. *Annu Rev Biochem* 2005;**74**:481–514.

Gosden R, Trasler J, Lucifero D, Faddy M. Rare congenital disorders, imprinted genes, and assisted reproductive technology. *Lancet* 2003; **361**:1975–1977.

Grimes SR Jr, Henderson N. Hyperacetylation of histone H4 in rat testis spermatids. *Exp Cell Res* 1984;**152**:91–97.

Hajkova P, Erhardt S, Lane N, Haaf T, El-Maarri O, Reik W, Walter J, Surani MA. Epigenetic reprogramming in mouse primordial germ cells. *Mech Dev* 2002;**117**:15–23.

Hammoud S, Emery BR, Aoki VW, Carrell DT. Identification of genetic variation in the 5' and 3' non-coding regions of the protamine genes in patients with protamine deregulation. *Arch Androl* 2007; **53**:267–274.

Hammoud S, Emery BR, Dunn D, Weiss RB, Carrell DT. Sequence alterations in the YBX2 gene are associated with male factor infertility. *Fertil Steril* 2009a;**91**:1090–1095.

Hammoud SS, Nix DA, Zhang H, Purwar J, Carrell DT, Cairns BR. Distinctive chromatin in human sperm packages genes for embryo development. *Nature* 2009b;**460**:473–478.

Hammoud S, Purwar J, Pflueger C, Cairns BR, Carrell DT. Alterations in sperm DNA methylation patterns at imprinted loci in two classes of infertility. *Fertil Steril* 2009; Epub ahead of print October 30.

Hansen M, Kurinczuk JJ, Bower C, Webb S. The risk of major birth defects after intracytoplasmic sperm injection and *in vitro* fertilization. *N Engl J Med* 2002;**346**:725–730.

Hata K, Okano M, Lei H, Li E. Dnmt3L cooperates with the Dnmt3 family of *de novo* DNA methyltransferases to establish maternal imprints in mice. *Development* 2002;**129**:1983–1993.

Hazzouri M, Pivot-Pajot C, Faure AK, Usson Y, Pelletier R, Sele B, Khochbin S, Rousseaux S. Regulated hyperacetylation of core histones during mouse spermatogenesis: involvement of histone deacetylases. *Eur J Cell Biol* 2000;**79**:950–960.

Houshdaran S, Cortessis VK, Siegmund K, Yang A, Laird PW, Sokol RZ. Widespread epigenetic abnormalities suggest a broad DNA methylation erasure defect in abnormal human sperm. *PLoS One* 2007; **2**:e1289.

Hoyer-Fender S. Molecular aspects of XY body formation. *Cytogenet Genome Res* 2003;**103**:245–255.

Hsieh CL. In vivo activity of murine *de novo* methyltransferases, Dnmt3a and Dnmt3b. *Mol Cell Biol* 1999;**19**:8211–8218.

Illingworth R, Kerr A, Desousa D, Jorgensen H, Ellis P, Stalker J, Jackson D, Clee C, Plumb R, Rogers J *et al*. A novel CpG island set identifies tissue-specific methylation at developmental gene loci. *PLoS Biol* 2008; **6**:e22.

Jaenisch R, Bird A. Epigenetic regulation of gene expression: how the genome integrates intrinsic and environmental signals. *Nat Genet* 2003;**33**:245–254.

Jaenisch R, Jahner D. Methylation, expression and chromosomal position of genes in mammals. *Biochim Biophys Acta* 1984;**782**:1–9.

Jenuwein T, Allis CD. Translating the histone code. *Science* 2001; **293**:1074–1080.

Jones PA, Baylin SB. The epigenomics of cancer. *Cell* 2007;**128**:683–692.

Kagami M, Nagai T, Fukami M, Yamazawa K, Ogata T. Silver-Russell syndrome in a girl born after *in vitro* fertilization: partial hypermethylation at the differentially methylated region of PEG1/ MEST. *J Assist Reprod Genet* 2007;**24**:131–136.

Kalra SK, Molinaro TA. The association of *in vitro* fertilization and perinatal morbidity. *Semin Reprod Med* 2008;**26**:423–435.

Kaneda M, Okano M, Hata K, Sado T, Tsujimoto N, Li E, Sasaki H. Essential role for *de novo* DNA methyltransferase Dnmt3a in paternal and maternal imprinting. *Nature* 2004;**429**:900–903.

Katz DJ, Edwards TM, Reinke V, Kelly WG. A *C. elegans* LSD1 demethylase contributes to germline immortality by reprogramming epigenetic memory. *Cell* 2009;**137**:308–320.

Khalil AM, Boyar FZ, Driscoll DJ. Dynamic histone modifications mark sex chromosome inactivation and reactivation during mammalian spermatogenesis. *Proc Natl Acad Sci USA* 2004;**101**:16583–16587.

Kimmins S, Sassone-Corsi P. Chromatin remodelling and epigenetic features of germ cells. *Nature* 2005;**434**:583–589.

Kobayashi H, Sato A, Otsu E, Hiura H, Tomatsu C, Utsunomiya T, Sasaki H, Yaegashi N, Arima T. Aberrant DNA methylation of imprinted loci in sperm from oligospermic patients. *Hum Mol Genet* 2007;**16**:2542–2551.

Kobayashi H, Hiura H, John RM, Sato A, Otsu E, Kobayashi N, Suzuki R, Suzuki F, Hayashi C, Utsunomiya T *et al*. DNA methylation errors at

Downloaded from http://molehr.oxfordjournals.org/ at Carleton University on June 22, 2014

imprinted loci after assisted conception originate in the parental sperm. *Eur J Hum Genet.* 2009; Epub ahead of print May 27.

Kopp JL, Ormsbee BD, Desler M, Rizzino A. Small increases in the level of Sox2 trigger the differentiation of mouse embryonic stem cells. *Stem Cells* 2008;**26**:903–911.

Kurtz K, Martinez-Soler F, Ausio J, Chiva M. Acetylation of histone H4 in complex structural transitions of spermiogenic chromatin. *J Cell Biochem* 2007;**102**:1432–1441.

Lachner M, Jenuwein T. The many faces of histone lysine methylation. *Curr Opin Cell Biol* 2002;**14**:286–298.

Lahn BT, Tang ZL, Zhou J, Barndt RJ, Parvinen M, Allis CD, Page DC. Previously uncharacterized histone acetyltransferases implicated in mammalian spermatogenesis. *Proc Natl Acad Sci USA* 2002;**99**: 8707–8712.

Laprise SL. Implications of epigenetics and genomic imprinting in assisted reproductive technologies. *Mol Reprod Dev* 2009;**76**:1006–1018.

Lee MG, Wynder C, Cooch N, Shiekhattar R. An essential role for CoREST in nucleosomal histone 3 lysine 4 demethylation. *Nature* 2005;**437**:432–435.

Lee TI, Jenner RG, Boyer LA, Guenther MG, Levine SS, Kumar RM, Chevalier B, Johnstone SE, Cole MF, Isono K *et al.* Control of developmental regulators by Polycomb in human embryonic stem cells. *Cell* 2006;**125**:301–313.

Lei H, Oh SP, Okano M, Juttermann R, Goss KA, Jaenisch R, Li E. *De novo* DNA cytosine methyltransferase activities in mouse embryonic stem cells. *Development* 1996;**122**:3195–3205.

Li E. Chromatin modification and epigenetic reprogramming in mammalian development. *Nat Rev Genet* 2002;**3**:662–673.

Li E, Bestor TH, Jaenisch R. Targeted mutation of the DNA methyltransferase gene results in embryonic lethality. *Cell* 1992; **69**:915–926.

Li E, Beard C, Forster AC, Bestor TH, Jaenisch R. DNA methylation, genomic imprinting, and mammalian development. *Cold Spring Harb Symp Quant Biol* 1993;**58**:297–305.

Lim D, Bowdin SC, Tee L, Kirby GA, Blair E, Fryer A, Lam W, Oley C, Cole T, Brueton LA *et al.* Clinical and molecular genetic features of Beckwith–Wiedemann syndrome associated with assisted reproductive technologies. *Hum Reprod* 2009;**24**:741–747.

Lu S, Xie YM, Li X, Luo J, Shi XQ, Hong X, Pan YH, Ma X. Mass spectrometry analysis of dynamic post-translational modifications of TH2B during spermatogenesis. *Mol Hum Reprod* 2009;**15**:373–378.

Lyko F, Ramsahoye BH, Kashevsky H, Tudor M, Mastrangelo MA, Orr-Weaver TL, Jaenisch R. Mammalian (cytosine-5) methyltransferases cause genomic DNA methylation and lethality in Drosophila. *Nat Genet* 1999;**23**:363–366.

Maher ER. Imprinting and assisted reproductive technology. *Hum Mol Genet* 2005;**14**:R133–R138.

Manipalviratn S, DeCherney A, Segars J. Imprinting disorders and assisted reproductive technology. *Fertil Steril* 2009;**91**:305–315.

Marques CJ, Carvalho F, Sousa M, Barros A. Genomic imprinting in disruptive spermatogenesis. *Lancet* 2004;**363**:1700–1702.

Marques CJ, Costa P, Vaz B, Carvalho F, Fernandes S, Barros A, Sousa M. Abnormal methylation of imprinted genes in human sperm is associated with oligozoospermia. *Mol Hum Reprod* 2008;**14**:67–74.

Matzuk MM, Lamb DJ. The biology of infertility: research advances and clinical challenges. *Nat Med* 2008;**14**:1197–1213.

McGraw S, Morin G, Vigneault C, Leclerc P, Sirard MA. Investigation of MYST4 histone acetyltransferase and its involvement in mammalian gametogenesis. *BMC Dev Biol* 2007;**7**:123.

Miyamoto T, Hasuike S, Yogev L, Maduro MR, Ishikawa M, Westphal H, Lamb DJ. Azoospermia in patients heterozygous for a mutation in SYCP3. *Lancet* 2003;**362**:1714–1719.

Mohn F, Weber M, Rebhan M, Roloff TC, Richter J, Stadler MB, Bibel M, Schubeler D. Lineage-specific polycomb targets and *de novo* DNA methylation define restriction and potential of neuronal progenitors. *Mol Cell* 2008;**30**:755–766.

Morgan HD, Sutherland HG, Martin DI, Whitelaw E. Epigenetic inheritance at the agouti locus in the mouse. *Nat Genet* 1999; **23**:314–318.

Morgan HD, Santos F, Green K, Dean W, Reik W. Epigenetic reprogramming in mammals. *Hum Mol Genet* 2005;**14**:R47–R58.

Muller J, Kassis JA. Polycomb response elements and targeting of Polycomb group proteins in Drosophila. *Curr Opin Genet Dev* 2006; **16**:476–484.

Nair M, Nagamori I, Sun P, Mishra DP, Rheaume C, Li B, Sassone-Corsi P, Dai X. Nuclear regulator Pygo2 controls spermiogenesis and histone H3 acetylation. *Dev Biol* 2008;**320**:446–455.

Ng HH, Bird A. DNA methylation and chromatin modification. *Curr Opin Genet Dev* 1999;**9**:158–163.

Oakes CC, La Salle S, Smiraglia DJ, Robaire B, Trasler JM. Developmental acquisition of genome-wide DNA methylation occurs prior to meiosis in male germ cells. *Dev Biol* 2007a;**307**:368–379.

Oakes CC, La Salle S, Smiraglia DJ, Robaire B, Trasler JM. A unique configuration of genome-wide DNA methylation patterns in the testis. *Proc Natl Acad Sci USA* 2007b;**104**:228–233.

O'Bryan MK, de Kretser D. Mouse models for genes involved in impaired spermatogenesis. *Int J Androl* 2006;**29**:76–89; discussion 105–108.

Okada Y, Scott G, Ray MK, Mishina Y, Zhang Y. Histone demethylase JHDM2A is critical for Tnp1 and Prm1 transcription and spermatogenesis. *Nature* 2007;**450**:119–123.

Okano M, Bell DW, Haber DA, Li E. DNA methyltransferases Dnmt3a and Dnmt3b are essential for *de novo* methylation and mammalian development. *Cell* 1999;**99**:247–257.

Oliva R. Protamines and male infertility. *Hum Reprod Update* 2006; **12**:417–435.

Ooi SK, Qiu C, Bernstein E, Li K, Jia D, Yang Z, Erdjument-Bromage H, Tempst P, Lin SP, Allis CD *et al.* DNMT3L connects unmethylated lysine 4 of histone H3 to *de novo* methylation of DNA. *Nature* 2007; **448**:714–717.

Oswald J, Engemann S, Lane N, Mayer W, Olek A, Fundele R, Dean W, Reik W, Walter J. Active demethylation of the paternal genome in the mouse zygote. *Curr Biol* 2000;**10**:475–478.

Panning B, Jaenisch R. DNA hypomethylation can activate Xist expression and silence X-linked genes. *Genes Dev* 1996;**10**:1991–2002.

Payne C, Braun RE. Histone lysine trimethylation exhibits a distinct perinuclear distribution in Plzf-expressing spermatogonia. *Dev Biol* 2006;**293**:461–472.

Pivot-Pajot C, Caron C, Govin J, Vion A, Rousseaux S, Khochbin S. Acetylation-dependent chromatin reorganization by BRDT, a testis-specific bromodomain-containing protein. *Mol Cell Biol* 2003; **23**:5354–5365.

Rakyan VK, Chong S, Champ ME, Cuthbert PC, Morgan HD, Luu KV, Whitelaw E. Transgenerational inheritance of epigenetic states at the murine Axin(Fu) allele occurs after maternal and paternal transmission. *Proc Natl Acad Sci USA* 2003;**100**:2538–2543.

Raman R, Narayan G. 5-Aza deoxyCytidine-induced inhibition of differentiation of spermatogonia into spermatocytes in the mouse. *Mol Reprod Dev* 1995;**42**:284–290.

Rangasamy D, Berven L, Ridgway P, Tremethick DJ. Pericentric heterochromatin becomes enriched with H2A.Z during early mammalian development. *EMBO J* 2003;**22**:1599–1607.

Reefhuis J, Honein MA, Schieve LA, Correa A, Hobbs CA, Rasmussen SA. Assisted reproductive technology and major structural birth defects in the United States. *Hum Reprod* 2009;**24**:360–366.

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Reik W, Dean W, Walter J. Epigenetic reprogramming in mammalian development. *Science* 2001;**293**:1089–1093.

Reik W, Santos F, Mitsuya K, Morgan H, Dean W. Epigenetic asymmetry in the mammalian zygote and early embryo: relationship to lineage commitment? *Philos Trans R Soc Lond B Biol Sci* 2003;**358**:1403–1409; discussion 1409.

Royo H, Bortolin ML, Seitz H, Cavaille J. Small non-coding RNAs and genomic imprinting. *Cytogenet Genome Res* 2006;**113**:99–108.

Schwartz YB, Kahn TG, Nix DA, Li XY, Bourgon R, Biggin M, Pirrotta V. Genome-wide analysis of Polycomb targets in Drosophila melanogaster. *Nat Genet* 2006;**38**:700–705.

Seif MM, Edi-Osagie EC, Farquhar C, Hooper L, Blake D, McGinlay P. Assisted hatching on assisted conception (IVF & ICSI). *Cochrane Database Syst Rev* 2006; CD001894.

Seifertova M, Vesely J, Cihak A. Enhanced mortality in offsprings of male mice treated with 5-azacytidine prior to mating. Morphological changes in testes. *Neoplasma* 1976;**23**:53–60.

Seitz H, Royo H, Bortolin ML, Lin SP, Ferguson-Smith AC, Cavaille J. A large imprinted microRNA gene cluster at the mouse Dlk1-Gtl2 domain. *Genome Res* 2004;**14**:1741–1748.

Seki Y, Hayashi K, Itoh K, Mizugaki M, Saitou M, Matsui Y. Extensive and orderly reprogramming of genome-wide chromatin modifications associated with specification and early development of germ cells in mice. *Dev Biol* 2005;**278**:440–458.

Seki Y, Yamaji M, Yabuta Y, Sano M, Shigeta M, Matsui Y, Saga Y, Tachibana M, Shinkai Y, Saitou M. Cellular dynamics associated with the genome-wide epigenetic reprogramming in migrating primordial germ cells in mice. *Development* 2007;**134**:2627–2638.

Shi Y, Lan F, Matson C, Mulligan P, Whetstine JR, Cole PA, Casero RA. Histone demethylation mediated by the nuclear amine oxidase homolog LSD1. *Cell* 2004;**119**:941–953.

Sonnack V, Failing K, Bergmann M, Steger K. Expression of hyperacetylated histone H4 during normal and impaired human spermatogenesis. *Andrologia* 2002;**34**:384–390.

Steel AJ, Sutcliffe A. Long-term health implications for children conceived by IVF/ICSI. *Hum Fertil (Camb)* 2009;**12**:21–27.

Suetake I, Shinozaki F, Miyagawa J, Takeshima H, Tajima S. DNMT3L stimulates the DNA methylation activity of Dnmt3a and Dnmt3b through a direct interaction. *J Biol Chem* 2004;**279**: 27816–27823.

Suganuma T, Workman JL. Crosstalk among Histone Modifications. *Cell* 2008;**135**:604–607.

Surani MA. Imprinting and the initiation of gene silencing in the germ line. *Cell* 1998;**93**:309–312.

Takahashi K, Tanabe K, Ohnuki M, Narita M, Ichisaka T, Tomoda K, Yamanaka S. Induction of pluripotent stem cells from adult human fibroblasts by defined factors. *Cell* 2007;**131**:861–872.

Tanay A, O'Donnell AH, Damelin M, Bestor TH. Hyperconserved CpG domains underlie Polycomb-binding sites. *Proc Natl Acad Sci USA* 2007;**104**:5521–5526.

Tanphaichitr N, Sobhon P, Taluppeth N, Chalermisarachai P. Basic nuclear proteins in testicular cells and ejaculated spermatozoa in man. *Exp Cell Res* 1978;**117**:347–356.

Trasler JM. Gamete imprinting: setting epigenetic patterns for the next generation. *Reprod Fertil Dev* 2006;**18**:63–69.

Walsh CP, Chaillet JR, Bestor TH. Transcription of IAP endogenous retroviruses is constrained by cytosine methylation. *Nat Genet* 1998; **20**:116–117.

Ward WS, Coffey DS. DNA packaging and organization in mammalian spermatozoa: comparison with somatic cells. *Biol Reprod* 1991; **44**:569–574.

Weber M, Hellmann I, Stadler MB, Ramos L, Paabo S, Rebhan M, Schubeler D. Distribution, silencing potential and evolutionary impact of promoter DNA methylation in the human genome. *Nat Genet* 2007;**39**:457–466.

Webster KE, O'Bryan MK, Fletcher S, Crewther PE, Aapola U, Craig J, Harrison DK, Aung H, Phutikanit N, Lyle R *et al*. Meiotic and epigenetic defects in Dnmt3L-knockout mouse spermatogenesis. *Proc Natl Acad Sci USA* 2005;**102**:4068–4073.

Wernig M, Meissner A, Foreman R, Brambrink T, Ku M, Hochedlinger K, Bernstein BE, Jaenisch R. *In vitro* reprogramming of fibroblasts into a pluripotent ES-cell-like state. *Nature* 2007;**448**:318–324.

Wykes SM, Krawetz SA. The structural organization of sperm chromatin. *J Biol Chem* 2003;**278**:29471–29477.

Zhu B, Zheng Y, Pham AD, Mandal SS, Erdjument-Bromage H, Tempst P, Reinberg D. Monoubiquitination of human histone H2B: the factors involved and their roles in HOX gene regulation. *Mol Cell* 2005;**20**:601–611.

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